nature chemistry

Article

https://doi.org/10.1038/s41557-025-01835-7

Bidentate *N*-ligand-assisted gold redox catalysis with hydrogen peroxide

Received: 10 October 2024

Accepted: 15 April 2025

Published online: 29 May 2025

Check for updates

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Gold redox catalysis, which exploits the ability of strong π-acid activation in combination with redox reactions, has emerged as an attractive synthetic method with unique reactivities compared to other transition metals. However, gold redox chemistry bears the challenge to overcome the high redox potential of Au(I)/Au(III) (1.41 V). The classical strategy of gold redox catalysis applies strong external chemical oxidants which inevitably results in low atom economy and substrate limitations due to incompatibility with functional groups. Here we report a bidentate *N*-ligand (for example, Phen, Bpy) assisted gold redox catalysis using H_2O_2 as oxidant, which proved to be generally applicable for many forms of coupling reactions. In addition, C(sp^2)–C(sp^2) bicyclization coupling (cross-coupling of two cyclized substrates) is accessible under our conditions. Mechanistic studies reveal a redox elimination process in which a bidentate *N*-ligand is crucial for the catalytic cycle. The formation of alkynyl-Au^{III}–OH and vinyl-Au^{III}–OH species is the key process for the synergistic π-bond activation and Au^{I} oxidation.

Driven by its strong π-acidity and the resulting ability to activate C–C multiple bonds, homogeneous gold catalysis has flourished for over two decades[1,2]. Classical gold catalysis involves the electrophilic activation of C–C π-bonds, while the oxidation state of gold remains unchanged throughout the catalytic cycle (Fig. 1a)[3–6]. In contrast, gold redox catalysis relies on the oxidation or oxidative addition of Au(I) species and reductive elimination of Au(III) intermediates (Fig. 1b)[7–24]. This interplay between an effective switch of the Au(I)/Au(III) oxidation states, combined with the strong π-activation ability, delivers attractive synthetic methods with unique reactivities, especially if coupled with π-acid activation reactions[9,25–28]. However, the oxidation of Au(I) complexes is thermodynamically challenging due to the high redox potential of the Au(I)/Au(III) pair (1.41 V), and in addition, geometric changes initiated by the oxidative addition of linear Au(I) complexes make these processes kinetically unfavourable[26,28,29]. Current strategies to overcome the kinetic challenges involve hemilabile ligand-assisted[12,30–33] or photosensitive[34,35] oxidative additions of Au(I) species. Alternatively, a direct oxidation of Au(I) to Au(III) by an external oxidant can be conducted, which is accompanied by often unique reactivities, especially for coupling reactions based on π-acid activation, forming a distinctive modern gold redox catalytic system that is different from that of other transition metals. For example, Au(I)/Au(III) catalytic cycles can be promoted by strong external chemical oxidants such as high-valent iodine reagents (PIDA[18–22,36], PBX[37], EBX[14,16,38–42], CpBX[43], etc.) or by electrophilic halogen reagents (for example, Selectfluor)[11,15,44,45], and by photochemical[23,24,46,47] and electrochemical[13,17,48–50] oxidation.

It has been reported that the Au^{III}–OAc intermediates generated by the oxidation of Au(I) by an external oxidant (PIDA), and using the catalyst IPrAu^{I}OH, are the key species for C–H bond activation[18–22,36,51]. Additionally, compared to Au–OCOR species, Au–OH species are more basic, which should enable more effective trapping of in situ generated protons and therefore prevent protodemetallation. However,

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https://doi.org/10.1038/s41557-025-01835-7

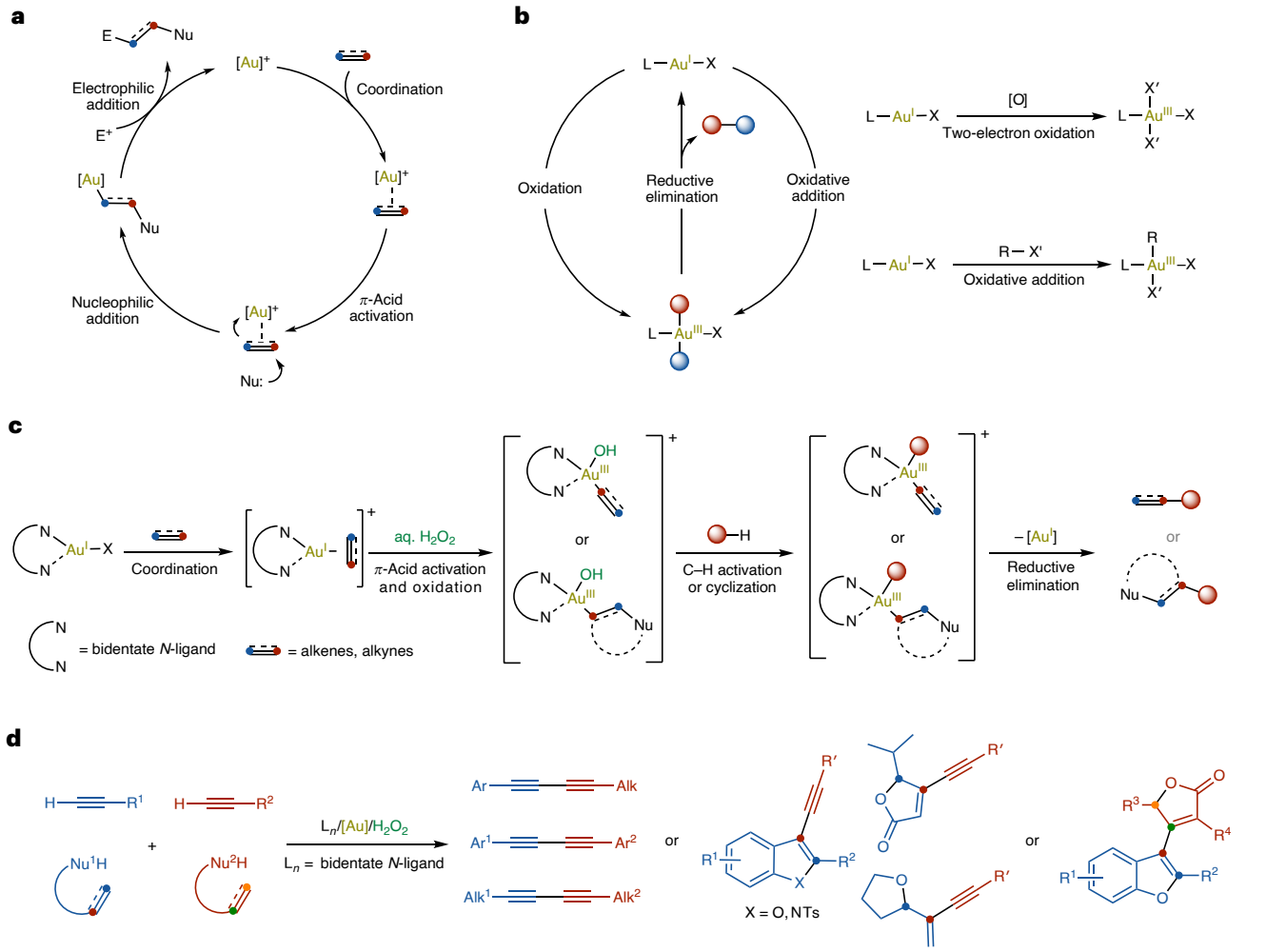


Fig. 1 | Gold redox catalysis and our strategy. a, General π-acid catalytic reactivity in gold catalysis. **b**, Gold redox catalysis and the challenges this faces, namely, a thermodynamic challenge due to the high redox potential of Au(I)/Au(III) (1.41 V), and a kinetic challenge caused by geometric changes in the linear Au(I) complexes. **c**, Our design: combining π-acid activation with an oxidation–reduction process using H_2O_2 as oxidant in gold redox catalysis. **d**, This work: bidentate *N*-ligand/Au/aq. H_2O_2 system for gold redox catalysed C(*sp*)–C(*sp*) cross-coupling, cyclization–alkynylation and bicyclization cross-coupling.

research on the participation of Au^{I}–OH/Au^{III}–OH species in catalytic redox cycles in homogeneous gold catalysis is still underdeveloped. The extremely low solubility of the square-planar tetrahydroxoaurate anion salt $[Au^{III}(OH)_4]^-$ makes it difficult to use this in homogeneous gold catalysis, and the methods for obtaining Au–OH species in situ are limited, and currently mainly involve ligand exchange of hydroxides[52,53]. In addition, Au–OH species, especially Au^{III}–OH species, are highly susceptible to dehydration, forming bridging oxide species (Au–O–Au) or being overoxidized to peroxo-gold species (Au–O–O–Au) under anhydrous conditions, which may result in a loss of catalytic activity[54].

Because the generation of Au–O–Au or Au–O–O–Au species from Au–OH species requires an anhydrous system, the development of gold redox catalysis in water-containing systems is the key to solving the above challenges. Hydrogen peroxide (H_2O_2), as an environmentally friendly and inexpensive oxidant in the form of an aqueous solution with high oxidation potential (+1.78 V)[55], provides a potential solution for gold redox catalysis in water-containing systems, avoiding the loss of gold catalytic activity due to the production of bridging oxide gold species from Au^{III}–OH species. Examples of gold redox catalysis with peroxides such as *tert*-butyl hydroperoxide (TBHP) as oxidants have been reported, which confirms that peroxides as oxidants have potential in gold redox catalysis[56–58]. However, water as a potential nucleophilic reagent readily undergoes nucleophilic addition reactions with gold-activated C–C π bonds[4], and gold intermediates such as vinyl-gold species are highly susceptible to protonation rather than further functionalized transformations in aqueous solutions[1,3,59], especially in the presence of H_2O_2 with its lower pK_a value, and therefore gold redox catalysis with aqueous H_2O_2 as oxidant remains challenging.

Interestingly, several studies have confirmed that bidentate *N*-ligands, such as bipyridine (Bpy) and 1,10-phenanthroline (Phen), enhance π back-donation from Au(I), thereby facilitating the oxidative addition progress[32,60]. Furthermore, for gold redox catalysis with external chemical oxidants, bidentate *N*-ligands can dramatically enhance the rate of reductive elimination of Au(III) complexes, particularly in the oxidative gold-catalysed C(*sp*)–C cross-coupling reaction[14,22,43]. Inspired by these studies, we conjectured that bidentate *N*-ligands (for example, Phen, Bpy) could enhance the oxidation rate of Au(I) species and the reactivity of Au(III) species for π-acid activation of alkynes, thus avoiding the potential side reactions in gold oxide catalysis. Based on this, we designed a bidentate *N*-ligand-assisted gold redox catalytic system that combines π-acid activation with an oxidation–reduction process using H_2O_2 as the oxidant (Fig. 1c). Predictably, the bidentate *N*-ligand can further enhance the activity of the Au(I) catalyst, making the Au(I) species more susceptible to coordination with C–C multiple bonds and activation of C–C π-bonds. The resulting Au(I) species is highly susceptible to oxidation by H_2O_2 to form the even more reactive Au^{III}–OH species, which can be exploited for further post-functionalization such as alkynylations and cyclizations. Here

we report a bidentate *N*-ligand/Au/H_2O_2 system for gold redox catalysis, which is suitable for many forms of coupling reactions (Fig. 1d). First, as a more attractive alternative to gold-catalysed C(*sp*)–C(*sp*) cross-couplings, this reaction system can be used not only for direct cross-couplings of different two different terminal alkynes (for example, aromatic with aliphatic alkynes), but also for the highly efficient cross-coupling of different alkynes with similar structural or electronic alkynes (for example, two different aromatic alkynes or two different aliphatic alkynes). Second, despite the very rapid rates of gold-catalysed redox homo-couplings or cross-couplings of terminal alkynes with chemical oxidants (for example, PIDA, Selectfluor)[11,22], this *N*-ligand/Au/H_2O_2 system can still be efficiently applied to a wide range of selective alkynylative cyclization reactions. Remarkably, this catalytic system can also provide a facile method for bicyclization cross-couplings between different cyclization coupling partners, which is not possible with existing gold catalytic strategies.

Results and discussion

Gold-catalysed C(*sp*)–C(*sp*) cross-coupling

Conjugated 1,3-diynes are attractive carbon frameworks, widely found in natural products, biologically active molecules and functional materials[61,62]. Typical methods of constructing 1,3-diynes include copper-catalysed Glaser–Hay coupling and Cadiot–Chodkiewicz coupling[63]. In recent years, Shi and co-workers developed gold-catalysed Glaser–Hay-type couplings with PIDA as external oxidant.[22] Liu and co-workers reported gold-catalysed Cadiot–Chodkiewicz-type cross-coupling with alkynyl hypervalent iodine reagents[14]. Despite these methods exhibiting good functional group compatibility and high selectivity compared with classical copper-catalysed C(*sp*)–C(*sp*) coupling, the use of PIDA and hypervalent iodine reagents brings low atom economy and high synthetic cost. In search of a green alternative strategy with high atom economy, we first expected that our bidentate *N*-ligand/Au/H_2O_2 system could be used for gold-catalysed cross-coupling of terminal alkynes.

To verify the feasibility of gold-catalysed redox cross-coupling with H_2O_2, we began by investigating the gold-catalysed C(*sp*)–C(*sp*) cross-coupling of terminal alkynes with 1-ethynyl-4-fluorobenzene **1a** and 2-methyl-3-butyn-2-ol **2a** as a test system. We checked a series of reaction parameters, including a set of gold catalysts, ligands, reaction temperatures, solvents, reaction times, equivalents of alkyne, combined equivalents of gold and ligands, and peroxide, to determine optimal reaction conditions (Supplementary Tables 1–7). Notably, homo-coupling is an unavoidable by-product in the cross-coupling reaction of different terminal alkynes, but due to the difference in the reaction rate of different alkynes with gold catalysts, especially for aromatic terminal alkynes and aliphatic terminal alkynes[22], the cross-coupling reaction remains the main product. Even if 1 equiv. of **2a** was used, the cross-coupling product could still be obtained in a good yield of 67% (Supplementary Table 7), but to improve the reaction efficiency, after screening the equivalents of **2a**, we decided to use 3 equiv. of **2** as the coupling partner. As shown in Table 1, the cross-coupling reactions of aromatic alkynes and aliphatic alkynes (**3a**–**3bd**), two aromatic alkynes (**4a**–**4q**), two aliphatic alkynes (**5a**–**5r**), and even complex derivatives of natural products or drugs (**6a**–**6i**), were carried out efficiently under standard conditions I, demonstrating the broad substrate range and functional group compatibility of this gold redox catalytic system. It is worth noting that this method is applicable to selectively cross-coupling two different alkynes with similar structural or electronic properties, for example, two aromatic or aliphatic alkynes, which are sometimes troublesome in classical cross-coupling methodologies. Among them, the structure of 1,3-diyne **3au** was confirmed by X-ray crystallographic analysis. The gramme-scale synthetic yields of the target product **3b** and **5r** were 90% and 74%, respectively, suggesting that scaling up the reaction to the gramme scale is readily achievable.

Gold-catalysed alkynylative cyclization

Compared to other transition metals, gold redox catalysis not only enables direct C–C cross-couplings, but also combines C–C π-bond activation with redox chemistry to construct more complex molecules of high synthetic value, in particular, functionalized molecules bearing attached cycles[3,4,64,65]. To verify the feasibility of this bidentate *N*-ligand/Au/H_2O_2 system for gold-catalysed redox π-activation–cyclization–functionalization, and to demonstrate that this catalytic system is not a simple alternative for conventional gold redox catalytic strategies with external chemical oxidants, we chose the alkynylative cyclization reaction of terminal alkynes as a template because this a reaction that cannot be achieved by conventional oxidants (for example, PIDA, Selectfluor).

We started our investigation of gold-catalysed alkynylative cyclization by using 2-(phenylethynyl)phenol **8a** and 1-ethynyl-4-fluorobenzene **1a** as a test system. Different reaction parameters were screened, including a series of ligands, gold catalysts, reaction times, equivalents of alkyne **1a**, oxidants, solvents, and combined equivalents of Au/phen/H_2O_2 (Supplementary Tables 8–14). Having established the conditions for this alkynylative cyclization, we set out to explore the scope of the nucleophilic partners and terminal alkynes (Table 2). First, we explored the scope of different terminal alkynes **1** to form 3-alkynylbenzofurans **9**, using 2-alkynylphenols **8** as the nucleophilic partners. Both (hetero)aromatic alkynes (**9a**–**9t**) and aliphatic alkynes (**9u**–**9ae**) are suitable substrates to obtain the target products in good to excellent yields. Among them, the structure of alkynylbenzofuran **9al** was confirmed by X-ray crystallographic analysis. Notably, upscaling the reaction to a gramme scale was easily possible, demonstrated by the synthesis of alkynylbenzofurans **3o** and **3v**, for which yields of 94% and 90% were achieved even with the use of DMSAuCl (1 mol%) and Phen (4 mol%), respectively. In addition, to underline the synthetic utility of the method, we next evaluated the potential to use this methodology to introduce benzofuran units into natural products and synthetic drugs. The target products (**9af**–**9ai**) were obtained in 53–91% yields even with 1.05 equiv. of complex alkynes as coupling partners.

Next, we explored the scope for a range of 2-alkynylphenols with different R^1 and R^2 groups (**9ak**–**9am**). It was quickly discovered that the R^2 groups introduced both aryl and alkyl functional groups efficiently (**9ak**–**9ax**). The R^1 group in the core of the 2-alkynylphenol partner (**9ay**–**9bc**) benzene introduces both electron-rich and electron-poor functional groups to obtain the target product efficiently. Because alkynes containing removable groups (for example, TIPS-, $OHMe_2C$-) can serve as handles for post-functionalization, we next investigated the reactivity of ethynyltriisopropylsilane and 2-methylbut-3-yn-2-ol as coupling partners. A range of 2-alkynylphenols with different functional groups (R^1 and R^2) also reacted well, furnished the target alkynylbenzofurans with yields ranging from 31% to 99% (**9bd**–**9bm**).

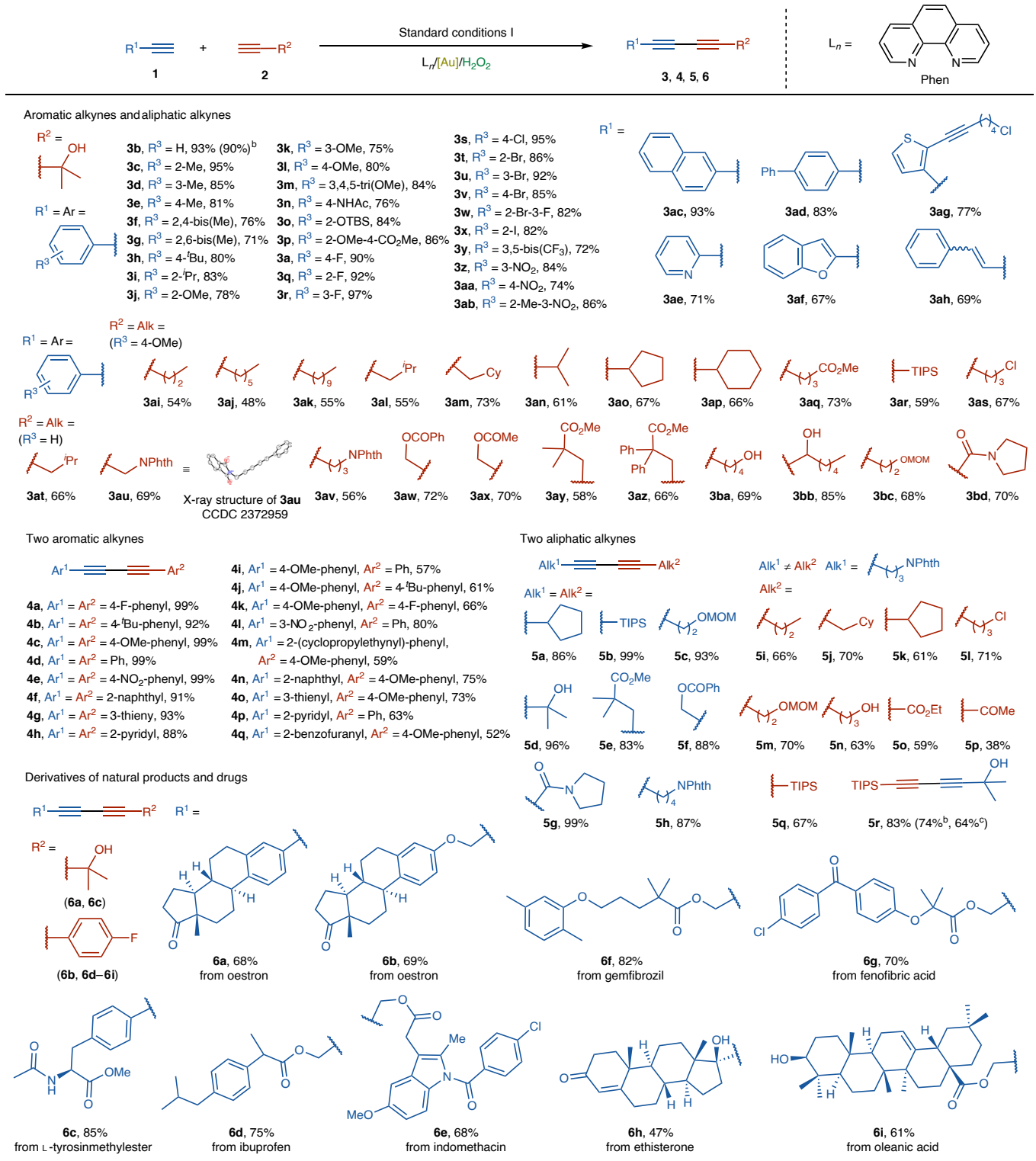
To further emphasize the general applicability of this bidentate *N*-ligand/Au/H_2O_2 system for gold-catalysed redox cyclization functionalization, other nucleophilic partners were also tested. 2-Alkynyl tosylaniline, *tert*-butyl dienoate and hexa-4,5-dien-1-ol as nucleophilic partners, under standard conditions, furnished the corresponding alkynylative cyclization products (**9bn**–**9bp**). In addition, we also tested other nucleophilic reagents, including nitrogen-, oxygen- and carbon-based nucleophilic reagents, but no alkynylative cyclization products were detected (Supplementary Fig. 3).

Gold-catalysed bicyclization cross-coupling

Gold-catalysed cyclization reactions are powerful methods for the synthesis of complex high-value carbo- and heterocyclic molecules. As an upgraded approach, bicyclization couplings, especially cyclization cross-coupling with two different cyclization coupling partners, can notably increase the synthetic efficiency because they

https://doi.org/10.1038/s41557-025-01835-7

Table 1 | Scope of the gold-catalysed C(*sp*)–C(*sp*) cross-coupling[a]

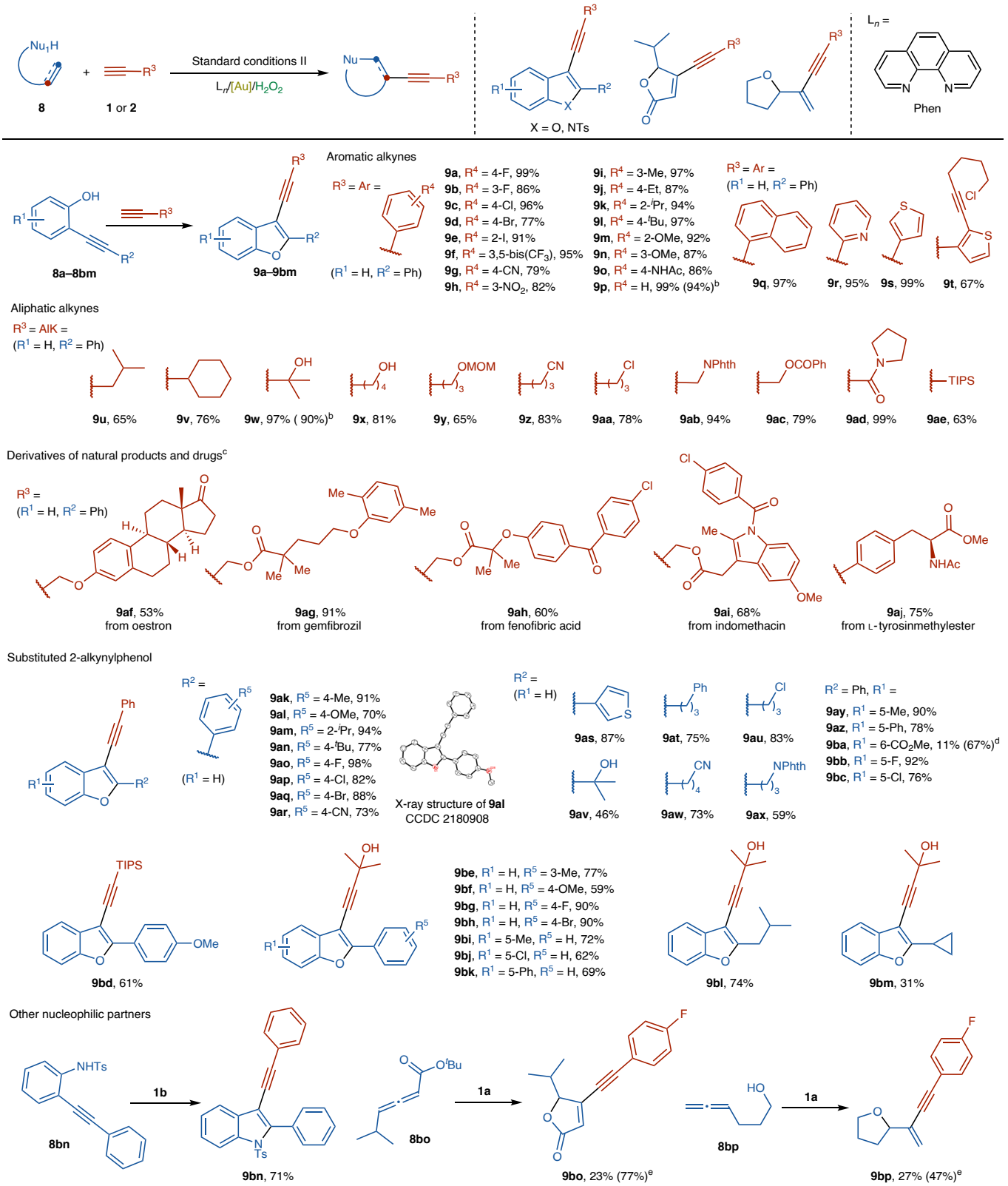


[a]Standard conditions I: alkyne **1** (0.2 mmol), alkyne **2** (0.6 mmol, 3 equiv.), Ph_3PAuCl (7.5 mol%), Phen (30 mol%) and H_2O_2 (1.6 mmol, 50 wt% in water) in MeCN (0.4 ml), 50 °C, 3 h. Isolated yields are reported. [b]Gramme-scale synthesis. [c]2-Methylbut-3-yn-2-ol (0.2 mmol, 1 equiv.), ethynyltriisopropylsilane (0.6 mmol, 3 equiv.). Alk, alkyne; iPr, *iso*-propyl; Cy, cyclohexyl; TIPS, triisopropylsilyl; NPhth, phthalimido; MOM, methoxymethyl.

provide additional possibilities for the construction of more complex molecules. However, the protonation of gold intermediates[1,59] and homo-coupling[56,66] as competing steps make the process more challenging, and so far there are no reports of gold-catalysed selective bicyclization/cross-coupling between two different types of cyclization coupling partners. To further emphasize the reactivity of our bidentate *N*-ligand/Au/H_2O_2 system, we explored the application of this system in more challenging bicyclization coupling reactions. We chose 2-(alkynyl) phenols and *tert*-butyl 2,3-dienoates as cyclization coupling partners to examine the feasibility of this approach.

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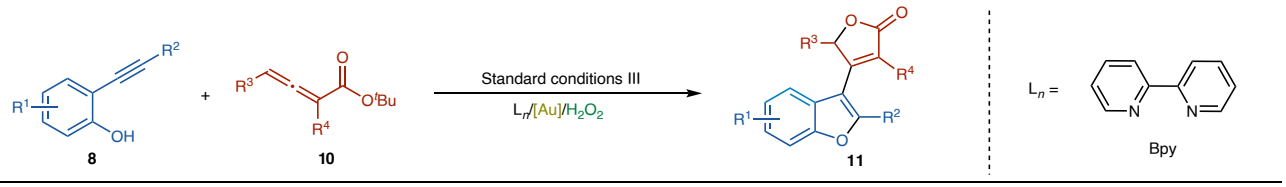
Table 2 | Scope of the gold-catalysed alkynylative cyclization[a]

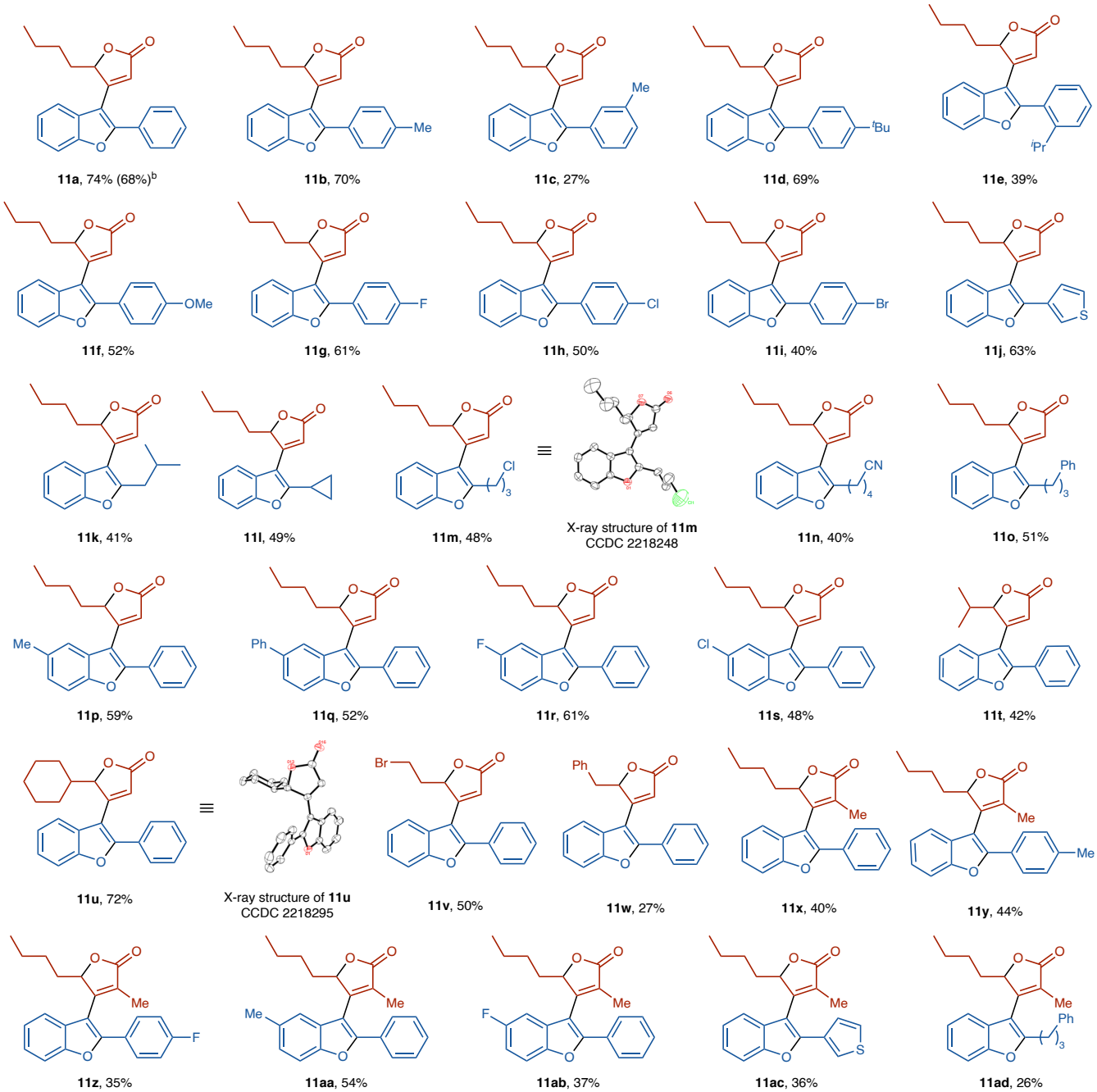


[a]Standard conditions II: nucleophilic partner (0.2 mmol), terminal alkyne (0.3 mmol, 1.5 equiv.), DMSAuCl (2.5 mol%), Phen (10 mol%) and H_2O_2 (1.6 mmol, 50 wt% in water) in methanol (0.4 ml), 50 °C. Isolated yields are reported. [b]Gramme-scale synthesis, DMSAuCl (1 mol%), Phen (4 mol%). [c]Alkyne **1** (0.21 mmol, 1.05 equiv.). [d]Recovered 2-alkynylphenol **1az**. [e]Alkyne **1** (0.4 mmol), DMSAuCl (7.5 mol%), Phen (30 mol%).

https://doi.org/10.1038/s41557-025-01835-7

Table 3 | Scope of the gold-catalysed bicyclization cross-coupling[a]





[a]Standard conditions III: 2-alkynylphenol **8** (0.2 mmol), *tert*-butyl dienoate **10** (0.3 mmol, 1.5 equiv.), DMSAuCl (5 mol%), Bpy (20 mol%) and H_2O_2 (1.6 mmol, 50 wt% in water) in methanol (0.4 ml, 0.5 M), 50 °C, overnight. Isolated yields are reported. [b]Gramme-scale synthesis. Bpy, 2,2′-bipyridine.

We started our investigation on gold-catalysed bicyclization/cross-coupling by using 2-(phenylethynyl)phenol **8a** and *tert*-butyl octa-2,3-dienoate **10a** as model substrates. Different reaction parameters were checked, including a series of ligands, gold catalysts, oxidants and solvents (Supplementary Tables 15–21). With the optimized conditions in hand, we explored the scope of substrates for the gold-catalysed bicyclization/cross-coupling (Table 3). We first focused on the scope of 2-alkynylphenols (**8**) bearing different R^2 groups as coupling partners. Aromatic substituents were evaluated first; both electron-donating and electron-withdrawing groups were tolerated

well, providing the corresponding alkenylbenzofurans in synthetically acceptable yields (**11a**–**11i**), and this reaction was also performed on a gramme scale in similarly good yield (**11a**). 2-Alkynylphenols bearing heteroaromatics (for example, thiophene) were also suitable substrates, giving the target product **11j** in 63% yield. 2-Alkynylphenols (**8**) bearing alkyl groups (R^2), including *iso*-butyl, cyclopropanyl and other alkyl-bearing functional groups, such as Cl-, CN- and Ph-, were all tolerated as well, affording the target products in moderate yields (**11k**–**11o**). Furthermore, we introduced electron-rich (Me-, Ph-) and electron-poor (F-, Cl-) functional groups into the R^1 groups of the benzene core of the 2-alkynylphenol partners; both types were tolerated well, furnishing the corresponding bicyclic coupling products (**11p**–**11s**) in moderate yields (48–61%). Next, our attention centred on the scope of *tert*-butyl dienoates (**10**) as coupling partners. We varied the R^3 groups by introducing alkyl groups with substantial steric bulk such as ^{i}Pr- and Cy- and alkyl groups bearing functional groups such as Br- and Ph-. All of these reacted smoothly, providing the target coupled products (**11t**–**11w**) in poor to good yields (27–72%). Among these, the structure of alkenylbenzofurans **11m** and **11u** was confirmed by X-ray crystallographic analysis of one of the isomers. Additionally, *tert*-butyl dienoates bearing another R^4 substituent (for example, Me-) were suitable substances as well, and the reactions with different 2-alkynylphenol partners delivered the target products in poor to moderate yield (**11x**–**11ad**).

Synthetic applications

This gold redox catalytic system can be readily applied to the synthesis of polyalkynes (Fig. 2a). For example, under standard conditions, treatment of the terminal alkynes **7a** and **7b**, which were synthesized from dimethylethanol and TIPS deprotection of **3b** and **5i**, respectively, afforded the tetrayne **7c** in good yield. As an alternative to the construction of polyalkyne by the cross-coupling of 1,3-diynes, we designed a relay synthesis strategy based on the consecutive use of commercially available compound 2-methyl-3-butyn-2-ol **2a** as coupling partner. The feasibility of this relay strategy was demonstrated by the effective construction of tetrayne **7c**. The structural modification of benzofuran as a ubiquitous skeleton of bioactive natural products and pharmaceuticals[67,68] remains an important field of research. The alkyne moiety introduced into the obtained products via gold-catalysed alkynylative cyclization offers an attractive synthetic handle for modifications, further multiplying the potential applications (Fig. 2b). Ethynylbenzofurans as valuable scaffolds were easily obtained by dimethylethanol deprotection of **9w**. Unsymmetric dibenzofuranoacetylene **12b** was efficiently prepared in 79% yield under standard conditions II. 3-Acetylbenzofuran **12c** was efficiently synthesized by water addition, using $AuCl_3$ as a catalyst. A $PtCl_2$-catalysed cyclization efficiently provided the heterotetracene **12d**. 1,3-Diynylbenzofuran **12e** was prepared in moderate yield via copper-catalysed Glaser–Hay coupling. In contrast, under our standard conditions I, 1,3-diynylbenzofuran **12f** was successfully obtained in good yield. Additionally, the 1,3-diyne group in the product offered a versatile handle for further derivatization (Fig. 2c). For example, furan (**12g**), thiophene (**12h**), pyrrole (**12k**) and pyridine (**12l**) derivatives could be readily obtained by the cycloaddition reaction of 1,3-diyne **12f** with different nucleophiles (for example, H_2O, Na_2S, $PhNH_2$ and $BnNH_2$). The treatment of **12f** with hydroxylamine hydrochloride ($NH_2OH{\cdot}HCl$) and hydrazine hydrate ($N_2H_4{\cdot}H_2O$), afforded 1*H*-pyrazole (**12i**) and isoxazole (**12j**), respectively. Among these products, the structures of thiophene (**12h**) and pyridine (**12l**) were confirmed by X-ray crystallographic analysis. To demonstrate the synthetic application of this gold-catalysed bicyclization coupling, a series of late-stage molecular modifications of compound **11a** were performed (Fig. 2d). The furan-2(*5H*)-one moiety on product **11a** could be transformed into a furan group (**12m**) by treatment with diisobutylaluminium hydride (DIBAL-H). Hydride reduction of **11a** with lithium aluminium hydride (LAH) afforded diol **12n**. The treatment of **11a** with trimethylsilyl trifluoromethanesulfonate (TMSOTf) and Et_3N led to furan-2(3*H*)-one **12o** in 79% yield. By using an excess of *n*-butyllithium, furan-2(5*H*)-one **11a** underwent ring opening to afford product **12p** in 68% yield. Furan-2(5*H*)-one **11a** reacted with an excess of alkynyl Grignard reagent to produce ethynylfuran **12q** via successive formate addition and hydroxyl elimination. Notably, under UVA (320–400 nm) light excitation, the furan-2(5*H*)-one moiety of **11a** underwent electrocyclic ring closure with the benzene ring (R^2) to furnish cyclized product **11r**. Selective hydrogenation of furan-2(5*H*)-one **11a** proceeded smoothly, giving product **12s** in excellent yield by treatment with Pd/C in methanol under 1 atm H_2 at room temperature. The hydrolysis of **11a** also proceeded smoothly, giving ring-opened acid **12t** in medium yield.

Mechanistic studies

To obtain insight into the operating mechanism of gold redox catalysis with H_2O_2, we conducted a detailed mechanistic study (Supplementary Figs. 4–30 and Supplementary Tables 22–26). In the case of gold-catalysed alkynylative cyclization, the reaction involves two pathways to produce vinyl-gold(III)[34,35,43,44,49] and alkynyl-gold(III)[11,14,16,17,21,22,39,69] intermediates, respectively. As shown in Fig. 3a, when DMSAuCl was added (without additives or with Phen, H_2O, H_2O_2 as additives alone), only trace amounts of **13a** were detected (entries 1–4). Under standard conditions II (in the absence of a terminal alkyne), the benzofuran **7a** was obtained in a yield of 92% after stirring for 24 h. In contrast, the replacement of H_2O_2 by H_2O was inefficient (entries 5 and 6). These results suggest that the catalytic process proceeds through a vinyl-gold(III) intermediate, and the rate of protonation is presumably much slower than the rate of alkynylation. Formation of gold(I) acetylide is a rapid process in gold catalysis, especially under neutral and weakly basic conditions[12–14,17,19,22]. Although our reaction system is acidic because the pH range of the H_2O_2 solution is 3–5, we cannot exclude whether the reaction involves acetylated gold intermediates, considering that we used 4 equiv. of ligands compared to gold, and that these pyridine-type ligands (for example, Phen, Bpy) may act as bases. With this in mind, we performed a set of control experiments using ((4-fluorophenyl)ethynyl)gold(I) **13e** and ((3-fluorophenyl)ethynyl)gold(I) **13f** (Supplementary Figs. 8–10). For example, as shown in Fig. 3b, under standard conditions II, stochiometric amounts of alkynyl-gold(I) **13e**, 2-(phenylethynyl)phenol **8a** and 1-ethynyl-3-fluorobenzene **1r** delivered an NMR yield of 78% for the product **9b** and only 6% for the product **9a**. Considering that the position of the substituent on the aromatic alkynes may have a slightly different binding affinity toward the gold cation, which may lead to different reactivities, a set of reverse cross-coupling experiments was carried out using ((3-fluorophenyl)ethynyl)gold(I) **13f**, 1-ethynyl-4-fluorobenzene **1a** and 2-(phenylethynyl)phenol **8a** as coupling partners (Supplementary Fig. 10). Under our standard conditions, the corresponding alkynyl cyclic products **9a** and **9b** were obtained in 60% and 14% NMR yields, respectively. This competition experiment shows that the in situ generation of alkynyl-gold(I) is unlikely (path A). To further understand the reaction mechanisms, we next used electrospray ionization mass spectrometry (ESI-MS) techniques to monitor the reaction mixtures; mass peaks for gold species **I**–**VIII** were all directly observed by MS (Fig. 3c and Supplementary Figs. 11–24). The formation of alkynyl-gold(III) species **III** and vinyl-gold(III) species **VI** may involve a synergistic process of π-bond activation and oxidation of Au(I) species. The key Au(III) species **VIII** connects the alkynylation or cyclization and reductive elimination steps. In this catalytic system, the weak coordination of the gold centre to the C–C π-bond is a crucial step, whether it is the oxidation of Au(I) species via H_2O_2 to form Au(III) species or the formation of Au(III) species **VIII** via alkynylation or cyclization.

With these preliminary mechanistic investigations in mind, our proposed catalytic cycle is presented in Fig. 4. The gold-catalysed alkynylative cyclization starts with a weak coordination of the Au(I) species **I** to the alkyne to produce Au(I) intermediates **II** or **V**, which

https://doi.org/10.1038/s41557-025-01835-7

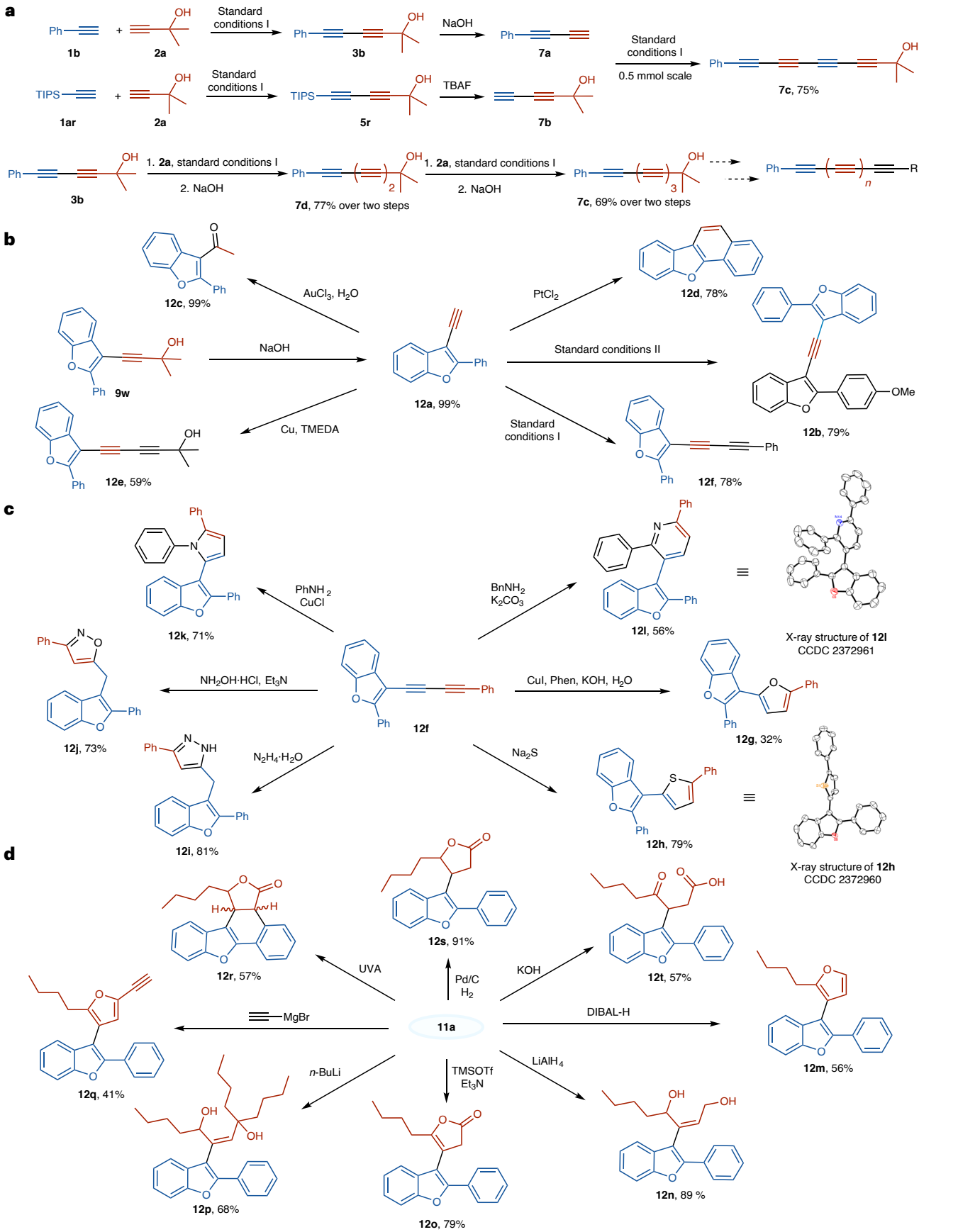


Fig. 2 | Comprehensive synthetic applications. a, Tetrayne synthesis and relay synthesis of polyynes under standard conditions I. **b**, Examples of post-functionalization modifications of acetylenic benzofurans as valuable scaffolds. TMEDA, *N,N,N′,N′*-tetramethylethylenediamine. **c**, Examples of post-functionalization modifications of 1,3-diyne skeletons as versatile handles. **d**, Examples of post-functionalization modifications of bicyclization cross-coupling products: synthetic transformation of furan-2(*5H*)-one skeleton.

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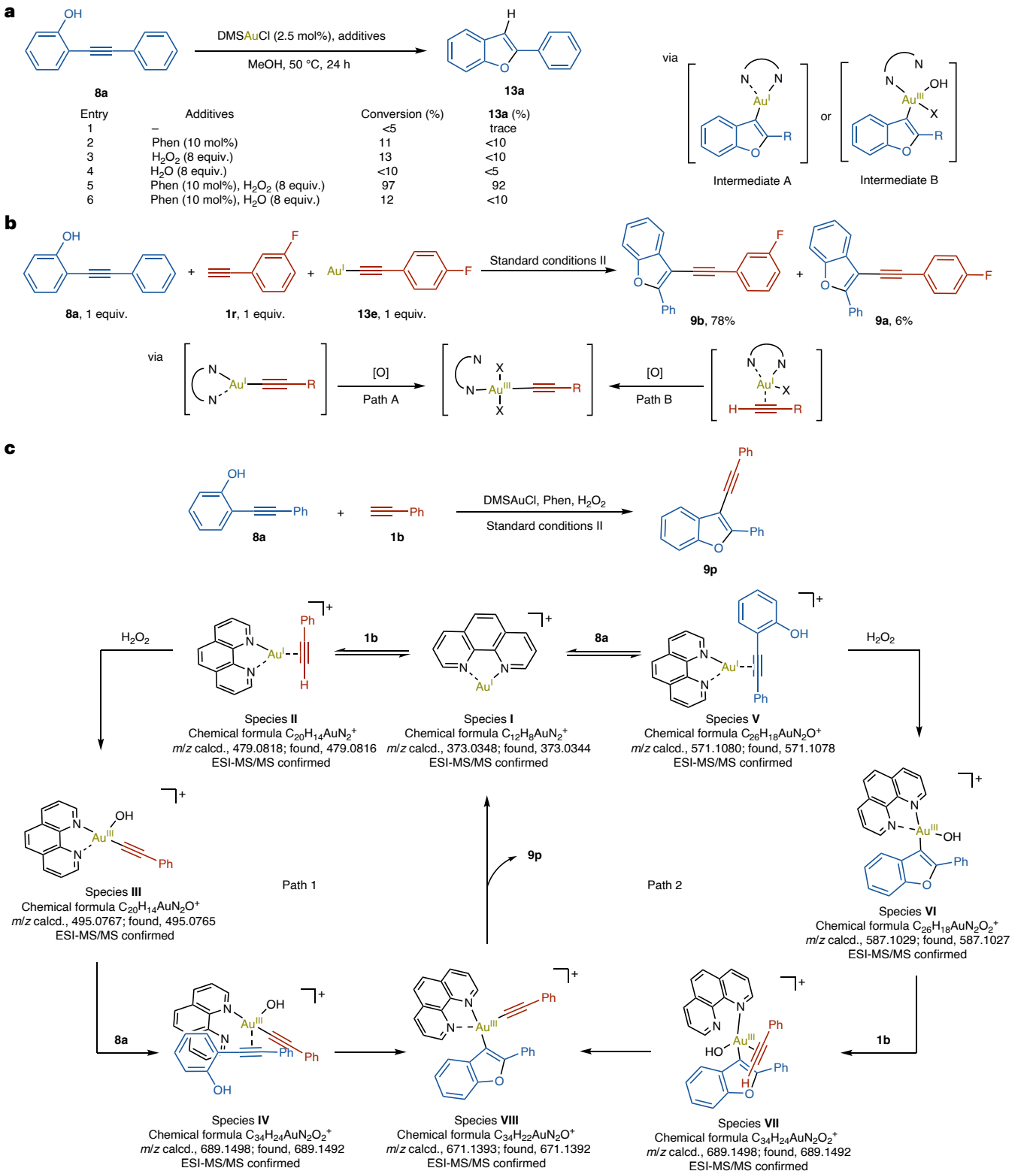


Fig. 3 | Mechanistic studies. a, Control experiments for the formation of the hydrogen-functionalized product. The catalytic process involves vinyl-gold(III) intermediate B rather than vinyl-gold(I) intermediate A. **b**, Competition experiments of alkynyl-gold(I) complex **13e** with terminal alkyne **1r**. Alkynylative cyclization gives **9b** as a major product. The reaction involves path B rather than path A. **c**, ESI-MS analysis of the gold-catalysed alkynylative cyclization under standard conditions II. The possible intermediates **I**–**VIII** involved in the reaction process were confirmed by ESI-MS. The reaction may involve both left and right catalytic cycles. Path 1 involves gold coordination with terminal alkynes, oxidation/alkynylation, coordination with 2-(phenylethynyl)phenol and cyclization processes. Path 2 involves gold coordination with 2-(phenylethynyl) phenol, oxidation/cyclization, coordination with terminal alkynes and alkynylation processes.

https://doi.org/10.1038/s41557-025-01835-7

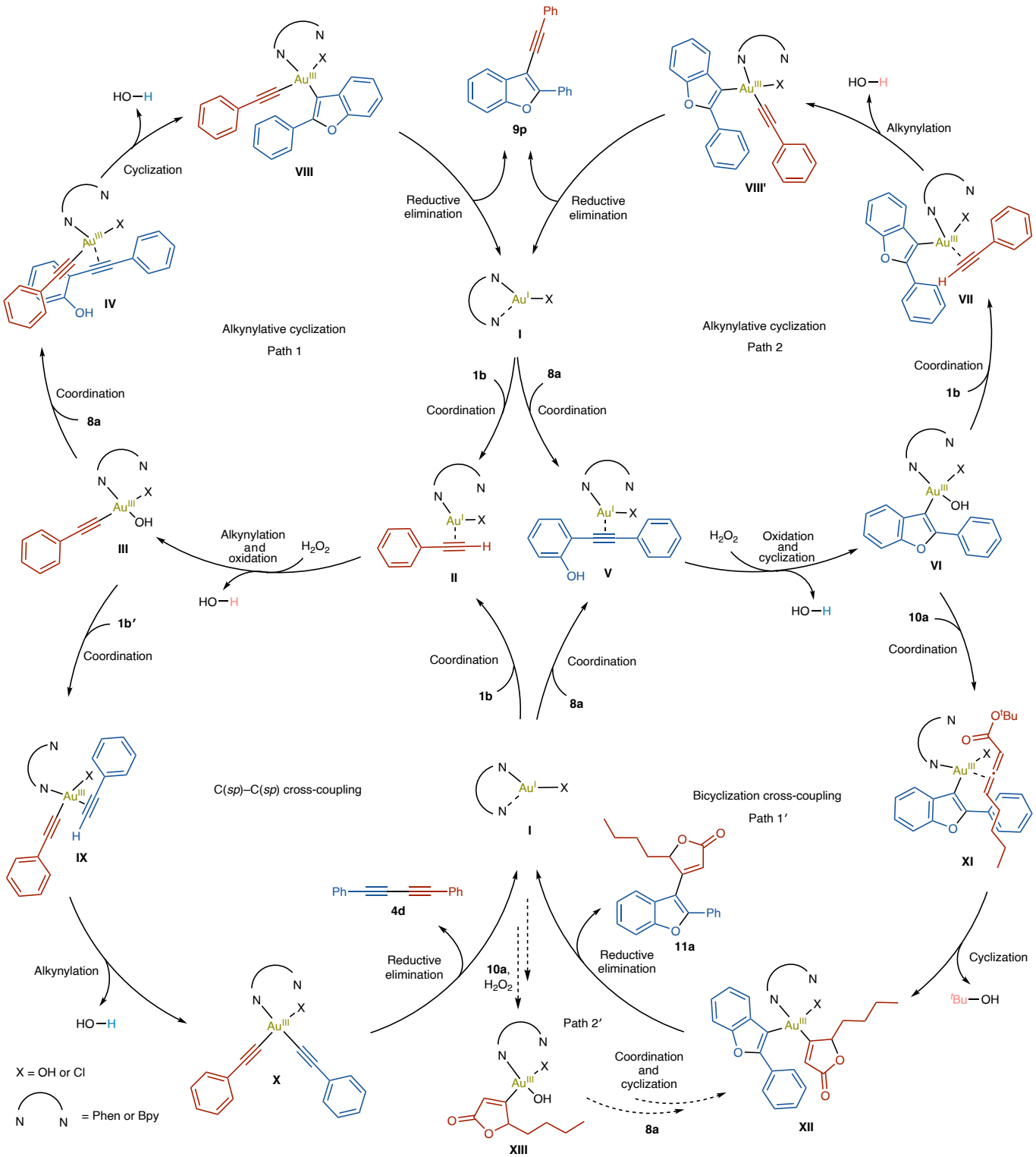


Fig. 4 | Plausible reaction pathway. The proposed catalytic cycle mechanisms for gold-catalysed C(*sp*)–C(*sp*) cross-couplings, alkynylative cyclization and bicyclization coupling all primarily involve coordination, oxidation, alkynylation or cyclization, and reductive elimination processes. The gold-catalysed alkynylative cyclization involves both the upper left and upper right catalytic cycles. Path 1 (upper left) starts with coordination of the Au(I) species **I** to the terminal alkyne to produce Au(I) intermediates **II**. Path 2 (upper right) starts with coordination of the Au(I) species **I** to 2-ethynylphenol to produce Au(I) intermediates **V**. The lower left catalytic cycle is the proposed catalytic cycle for gold-catalysed C(*sp*)–C(*sp*) cross-coupling. The lower right catalytic cycle (path 1′) is the proposed catalytic cycle for gold-catalysed bicyclization coupling, starting with the coordination of Au(I) species **I** to the cyclization partner **8a**, followed by a cyclization/oxidation reaction to give Au(III) intermediates **VI**. Path 2′ starts with the coordination of Au(I) species **I** to the cyclization partner **10a**, followed by a cyclization–oxidation reaction to give Au(III) intermediates **XIII**.

are then oxidized by H_2O_2 to produce alkynyl-gold(III) species **III** or vinyl-gold(III) species **VI**, respectively. Then the Au(III) centre is coordinated to the alkynes to produce the Au(III) species **IV** or **VII**, followed by alkynylation or cyclization to afford the Au(III) species **VIII**. Reductive elimination from intermediate **VIII** then affords the alkynylation products **9** under regeneration of the catalyst. For the gold-catalysed C(*sp*)–C(*sp*) cross-coupling, Au(III) species **III**, formed from Au(I) species **I**, provides Au(III) intermediate **IX** via coordination with another molecule of alkyne, which subsequently undergoes alkynylation to form Au(III) species **X**, followed by reductive elimination to form 1,3-diynes. It is noteworthy that gold species **IX** and **X** were also confirmed by ESI-MS (Supplementary Figs. 23 and 24). Different terminal alkynes (for example, aromatic alkynes, aliphatic alkynes) form Au(III) species **III** at different rates, but the selectivity of the reactions is indistinguishable. For the gold-catalysed bicyclization cross-coupling (for mechanistic studies, see Supplementary Figs. 25–30), Au(I) species **I** undergoes cyclization and oxidation processes to give Au(III) species **VI**. A weak coordination of the Au(III) centre to another cyclization partner produces Au(III) species **XI**, which then undergoes a secondary cyclization to give Au(III) species **XII**, followed by reductive elimination to furnish the bicyclization coupling product **11**. Furthermore, Au(I) species **I** and another cyclization partner (for example, *tert*-butyl dienoate) can also undergo cyclization and oxidation processes to produce Au(III) species **VIII**, which then undergoes similar process to afford the same Au(III) species **XII**.

Conclusion

We have developed a bidentate *N*-ligand/Au/H_2O_2 system for gold redox catalysis that is not only a more attractive substitute for common methodologies but also offers a unique tool for gold catalysis. Simple available bidentate *N*-ligands and gold catalysts combined with a green and inexpensive oxidant enable this gold redox catalytic system to be suitable for many forms of coupling reactions, including C(sp)–C(sp) cross-coupling, C(sp^2)–C(sp) cyclization coupling and C(sp^2)–C(sp^2) bicyclization coupling. The advantages of high atom economy, low synthetic costs, mild reaction conditions, remarkable selectivity, wide substrate range and good functional group tolerance demonstrate the great potential of this catalytic system for gold redox catalysis. A facile gramme-scale synthesis and a rich variety of synthetic transformations confirm the promising synthetic applications of these gold-catalysed coupling reactions. Several control experiments and ESI-MS techniques show that the reaction involves the Au(I)/Au(III) catalytic cycle, alkynyl-gold(III)–OH species **III** and vinyl-gold(III)–OH species **VI** as key intermediates, and bidentate *N*-ligands (for example, Phen) are crucial for the catalytic cycle. We anticipate a wide range of exciting applications for this bidentate ligand-promoted gold redox catalysis with H_2O_2 for C–C couplings and cyclization couplings, and for the π-bond activation and oxidation synergistic mechanism that underpins it.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41557-025-01835-7.

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Methods

General procedure for the gold-catalysed C(*sp*)–C(*sp*) cross-coupling of terminal alkynes (standard conditions I)

A 4-ml vial equipped with a magnetic stir bar was charged with Ph_3PAuCl (7.4 mg, 0.015 mmol, 7.5 mol%), Phen (10.8 mg, 0.06 mmol, 30 mol%), terminal alkynes **1** (0.2 mmol, if as solid), terminal alkynes **2** (0.6 mmol, 3 equiv., if as solid) in MeCN (0.4 ml), then terminal alkynes **1** (0.2 mmol, if as oil), terminal alkynes **2** (0.6 mmol, 3 equiv., if as oil) and H_2O_2 (89.6 µl, 1.6 mmol, 50 wt% in water) were added. The mixture was stirred at 50 °C for 3 h. After completion of the reaction (monitored by thin-layer chromatography (TLC); if the reaction has not completed, the reaction time is extended), the resulting solution was cooled to room temperature. Next, 2 ml of dichloromethane (DCM) was added, and the solution was dried over anhydrous Na_2SO_4, filtered and concentrated in vacuo. The residual mixture was purified by column chromatography on silica gel, to afford the products **3**.

General procedure for the gold-catalysed alkynylative cyclization (standard conditions II)

A 4-ml vial equipped with a magnetic stir bar was charged with DMSAuCl (1.5 mg, 0.005 mmol, 2.5 mol%), Phen (3.6 mg, 0.02 mmol,10 mol%), 2-(alkynyl)phenol **8** (0.2 mmol, if as solid), terminal alkynes **1** (0.3 mmol, 1.5 equiv., if as solid) in methanol (0.4 ml), then 2-(alkynyl)phenol **8** (0.2 mmol, if as oil), terminal alkynes **1** (0.3 mmol, 1.5 equiv., if as oil) and H_2O_2 (89.6 µl, 1.6 mmol, 50 wt% in water) were added. The mixture was stirred at 50 °C. After completion of the reaction (monitored by TLC; if the reaction has not completed, the reaction time is extended), the resulting solution was cooled to room temperature. Next, 2 ml DCM was added, and the solution was dried over anhydrous Na_2SO_4, filtered and concentrated in vacuo. The residual mixture was purified by column chromatography on silica gel, to afford the products **9**.

General procedure for the gold-catalysed bicyclization cross-coupling (standard conditions III)

A 4-ml vial equipped with a magnetic stir bar was charged with DMSAuCl (2.9 mg, 0.01 mmol, 5 mol%), Bpy (6.2 mg, 0.04 mmol, 20 mol%), 2-(alkynyl)phenol **8** (0.2 mmol, if as solid), *tert*-butyl dienoate **10** (0.3 mmol, 1.5 equiv., if as solid) in methanol (0.4 ml), then 2-(alkynyl)phenol **8** (0.2 mmol, if as oil), *tert*-butyl dienoate **10** (0.3 mmol, 1.5 equiv., if as oil) and H_2O_2 (89.6 µl, 1.6 mmol, 50 wt% in water) were added. The mixture was stirred at 50 °C overnight. After completion of the reaction (monitored by TLC; if the reaction has not completed, the reaction time is extended), the resulting solution was cooled to room temperature and then 2 ml DCM was added. The solution was dried over anhydrous Na_2SO_4, filtered and concentrated in vacuo. The residual mixture was purified by column chromatography or TLC on silica gel to afford the products **11**.

Data availability

The data supporting the findings of this work are provided in the Supplementary Information, including experimental procedures and characterization of new compounds. Crystallographic data for the structures reported in this Article have been deposited at the Cambridge Crystallographic Data Centre, under deposition numbers CCDC 2372959 (**3au**), CCDC 2180908 (**9al**), CCDC 2218248 (**11m**), CCDC 2218295 (**11u**), CCDC 2372960 (**12h**) and CCDC 2372961 (**12l**). Copies of the data can be obtained free of charge via https://www.ccdc.cam.ac.uk/structures/.

Acknowledgements

H.S., Y.T., Y.L. and T.W. are grateful for PhD fellowships from the China Scholarship Council (CSC). H.S. gratefully acknowledges the CPSF (number GZC20230165) and the NSFC (number 22401010). J.L. gratefully acknowledges the Alexander von Humboldt Foundation for a postdoctoral fellowship. H.H., P.M.S. and A.S.K.H. gratefully acknowledge the Hector Fellow Academy for the generous provision of funding. H.S. and N.J. acknowledge the NSFC (numbers 22293014, 22131002, 22161142019), the National Key R&D Program of China (number 2021YFA1501700), Changping Laboratory, the New Cornerstone Science Foundation through the New Cornerstone Investigator Program and the XPLORER PRIZE for financial support. We thank X. Zhang for his help with the ESI-MS mechanism experiments.

Author contributions

H.S. conceived and designed the experiments. A.S.K.H. and N.J. directed the research. H.S. carried out most of the experiments. H.S., M.R., J.L., Y.T., M.C.D., H.A., Y.L., T.W., H.H., P.M.S., N.J. and A.S.K.H. participated in discussion. P.K. analysed IR data. F.R. and T.O. conducted X-ray crystal structure analysis. H.S., M.R., N.J. and A.S.K.H. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41557-025-01835-7.

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Peer review information *Nature Chemistry* thanks Nitin Patil, Xiaodong Shi and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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